HS4002: Lecture 11B

Panel Data

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Class admin

What's remaining

1. Group presentation #2(Week 12)
2. In-class test 2 (Week 13)
3. Submission of team research paper (Week 14)

Help me with class design

1. About AI tools.
   - → More integration, or less? How did it affect your own learning?
     - To what extent would you prefer if I tried to restrict all AI-use for this course?
   - → More prescriptive or less? Do I tell you what/how to use, or let you figure it out on your own?
   - → More surveillance or less? Should I be more explicit in surveilling where students should and shouldn't use AI?

What's Panel Data?

Panel data

- Refers to data sets where we follow a sample across time.
  - $\rightarrow$ These samples can comprise individuals, families, firms, countries etc.
- We ask the *same members* of the sample the questions across different points in time.

What's Panel Data?

Entity	Year	Y	X1	X2	X3	.....
1	1	#	#	#	#	.....
1	2	#	#	#	#	.....
1	3	#	#	#	#	.....
:	:	:	:	:	:	:
2	1	#	#	#	#	.....
2	2	#	#	#	#	.....
2	3	#	#	#	#	.....
:	:	:	:	:	:	:
3	1	#	#	#	#	.....
3	2	#	#	#	#	.....
3	3	#	#	#	#	.....

Dealing with panel data

- We can continue using OLS regression.
- This is known as 'pooled OLS regression.'
- We interpret results similarly to what we did before.
- Problems:
  - $\rightarrow$ Serial correlation in errors (same person/unit is appearing).
  - $\rightarrow$ Omitted variable bias.

Panel models

We can use panel models that are better at exploiting unique features of panel data to our advantage. Most importantly, they help us address time-invariant confounders.

Omitted Variable Bias

- Recall omission of important variables can produce bias in our results.
  - $\rightarrow$ This is one of the key Gauss-Markov assumptions about strict exogeneity of the error term.
- Omitted variable bias can lead us to misleading conclusions:
  - $\rightarrow$ e.g. no relationship when there *is* a relationship, or the converse.

Example: the Efficacy of 'Prestige' Tuition.

- Suppose we are interested in the efficacy of attending 'prestige' private tuition (such as the Learning Lab) on educational outcomes.
- We go to a secondary school, and survey sec 1 students. We collect information on their PSLE grades and their tuition attendance, and regress tuition attendance on PSLE scores.

  $\rightarrow \; Y_{\text{PSLE, i}} = \beta 0 + \beta_T X_{T,i} + \epsilon_i$
- To what extent does β_{T} capture the effect of tuition?

Revisiting the Strict Exogeneity Assumption

- The math behind OLS regression makes the assumption that $E(\epsilon_i|X) = 0$, i.e. that any variance unexplained by our model is NOT related to the taking of tuition.
- What is ϵ_i? It's all possible attributes of a student i that we are unable to measure or include in our model.

Revisiting the Strict Exogeneity Assumption

$$Y_{\text{PSLE, i}} = \beta 0 + \beta_T X_{T,i} + \epsilon_i$$

- In simpler terms: all of these 'missing information' must be UNCORRELATED with tuition attendance.
  - $\rightarrow$ ϵ_i cannot influence the likelihood of attending tuition.
  - $\rightarrow$ What is ϵ_i? It's all possible attributes of a student i that we are unable to measure or include in our model.

Effects of omitting variables

- Suppose there are variables that we did not measure that may be actually highly correlated with prestige tuition attendance: (a) student ability (A) and (b) parental education-levels (PE).

Effects of omitting variables

- If $\beta_A > 0$ (i.e. higher ability predicts higher PSLE scores) and $Cov(T, A) < 0$ (i.e. higher ability = less likely to attend tuition):
  - $\rightarrow$ Then omitting A results in an underestimate of β_T – because more 'weak' students attend tuition, the actual effects of tuition are suppressed.

Effects of omitting variables

- If $\beta_P > 0$ (i.e. higher parents education predicts higher PSLE scores) and $Cov(P, T) > 0$ (i.e. those with more education are more likely to send kids for tuition):
  - $\rightarrow$ Then omitting P results in an overestimate of β_T – because more children from 'well-educated' parents attending prestige tuition, its effects are exaggerated.

How do we deal with omitted variable bias?

One way is to try to control for more! Measure more, and so on.

How do we deal with omitted variable bias?

But there's always a limit to what we can measure. There's actually a whole field in statistics/social-science dealing with exactly this (*causal inference*).

How panel data can help.

- We can think of omitted variables as two types:
  - $\rightarrow$ Time-variant variables that change over time.
  - $\rightarrow$ Time-invariant variables that do not change over time.
- The models that we learn today can help us account for time-invariant differences.

Fixed Effects Models

Building Intuition from "first differencing"

Suppose we are interested in modeling a variable y that has been measured across two time periods, t =1 and t =2.

Let y_{it} be the dependent variable of interest for person i at time t, such that,

$$y_{it} = \beta_0 + \delta_0 d_t + \beta_1 x_{it} + a_i + u_{it}$$

where d_t is a binary indicator for period, a_i captures all unobserved time-constant factors that affect the outcome, and u_{it} captures time-varying errors.

Building Intuition from "first differencing"

Panel models lets us eliminating time-invariant factors (a_i) entirely.

Building Intuition from "first differencing"

$$y_{i1} = \beta_o + \beta_1 x_{i,1} + a_i + u_{i1}$$

$$y_{i2} = (\beta_o + \delta_0 d_t) + \beta_1 x_{i,2} + a_i + u_{i2}$$

If we start differencing, then

$$(y_{i2} - y_{i1}) = \delta_o + \beta_1 (x_{i2} - x_{i1}) + (u_{i2} - u_{i1})$$

$$\Delta y_i = \delta_0 + \beta_1 \delta x_i + \Delta u_i$$

Notice a_i is now eliminated!

Example of tuition

Let $Y_{i,t}$ be a student's i English grade at semester t,

$$y_{it} = \beta_o + \beta_x X_{i,t} + P_i + A_i + u_{i1}$$

P_i represents their parents' education level, and A_i their innate aptitude at English. We do not have data on either of these. $X_{i,T}$ is a binary variable that tells us whether they had tuition at a time t. Over time, some students start to take tuition, or to stop taking tuition.

Example of tuition

A student's i English grade at semester 1 and 2 can be represented by these respectively.

$$y_{i1} = \beta_o + \beta_x X_{i,T} + P_i + A_i + u_{i1}$$

$$y_{i2} = (\beta_o + \delta_0 d_t) + \beta_x X_{i,T} + P_i + A_i + u_{i2}$$

$\delta_0 d_t$ are time-fixed effects included to account for changes over time.

Example of tuition

If we start differencing, then

$$(y_{i2}-y_{i1}) = (\beta_0-\beta_0)+\delta_o+\beta_x(x_{i2}-x_{i1})+(P_i-P_i)+(A_i-A_i)+(u_{i2}-u_{i1})$$

$$\Delta y_i = \delta_0 + \beta_1 \delta x_i + \Delta u_i$$

Time-invariant sources of variation such as student ability or parental background are now eliminated!

The “within" transformation

What we covered above is called the "first-differencing." Fixed effects models use the “within transformation" to accomplish what we described above. Sometime this transformation is also referred to as the “fixed effects transformation."

The “within" transformation

For each unit i,

$$y_{it} = \beta_1 x_{it1} + \beta_2 x_{it2} ... + \beta_k x_{itk} + a_i + u_{it}.$$

Then, average this equation over time T.

$$\bar{y}_i = \beta_1 \bar{x}_{it1} + \beta_2 \bar{x}_{it2} ... + \beta_k \bar{x}_{itk} a_i + \bar{u}_i$$

The “within" transformation

We then create time-demeaned data:

$$y_{it} - \bar{y}_i = \beta_1(x_{it1} - \bar{x}_{i1}) + ... + \beta_k(x_{itk} - \bar{x}_{ik}) + u_{it} - \bar{u}_i.$$

$$\ddot{y}_{it} = \beta_1\ddot{x}_{it1} + \beta_2\ddot{x}_{it2} + ... + \beta_k\ddot{x}_{itk} + \ddot{u}_{it}$$

Fixed Effects Estimation

- We then use the fixed effects, or “within," estimator to produce our model estimators.
- It’s called the “within" estimator because it uses the variation in y and x *within* each cross-sectional observation over time.
- One key assumption made is that the error term u_{it} is uncorrelated with regressors (independent variables) across all time periods.

Interpreting Results

- For a given individual (insert unit-individual as appropriate), a one-unit change in X over unit time is expected to produce a [BLANK] unit change in Y, ceteris paribus.
- Similar to OLS, but note that interpretations are about changes over time for a given individual.

Random Effects Models

Random Effects Models

Random effects models account for a_i in a different way. Unlike FE models, RE models assume that the unobserved time-invariant factors a_i are uncorrelated with each explanatory variable, i.e. $\text{Cov}(x_{itj}, a_i) = 0$.

Random Effects Models

As before, begin with

$$y_{it} = \beta_0 + \beta_1 x_{it1} + ...\beta_k x_{itk} + a_i + u_{it}.$$

We define the composite error term v_{it} as thus:

$$v_{it} = a_i + u_{it}$$

Random Effects Models

Define

$$\theta = 1 - \sqrt{\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_a^2}}.$$

Random Effects Models

Then, create quasi-demeaned data as thus:

$$y_{it} - \theta \bar{y}_i = \beta_0(1-\theta) + \beta_1(x_{it1} - \theta \bar{x}_{i1}) + \ldots + \beta_k(x_{itk} - \theta \bar{x}_{ik}) + (v_{it} - \theta \bar{v}_i)$$

The random effects transformation subtracts a fraction of that time average, where the fraction θ depends on σ_u, σ_a, and the number of time periods, T.

Why RE?

- Can estimate effects of time-invariant variables
  - $\rightarrow$ RE subtracts only a fraction of the individual mean, so a time-invariant regressor becomes $(1-\theta)a_i$ rather than zero.
  - $\rightarrow$ Since RE assumes α_i is uncorrelated with a_i, differences in y across individuals with different a_i can be attributed to a_i
- More efficient since it uses both within- and between-unit variation
- However, it relies on stronger statistical assumptions that need to be justified.

Interpreting Results

- Similar to OLS.
- Beta coefficients indicate the change in Y when the independent variables change one unit over time and across entities (average effect).
  - $\rightarrow$ e.g. On average across all time periods and individuals, a one unit-change in X produces a [BLANK] change in Y, ceteris paribus.

Diagnostic Tests

Lagrange Multiplier (LM) Diagnostics

- We can use LM tests to test for the presence of time-invariant unobservables (i.e. test for the necessity of FE models over pooled OLS).
- The null hypothesis is that there are no individual fixed effects.
- If we reject the null, we are saying that there are fixed effects present in the model.

  $\rightarrow$ i.e. if p value is small, opt for FE over OLS models.

Hausman Tests

- The Hausman test checks if the individual characteristics are correlated with the regressors.
- The null hypothesis is that the time-invariant individual characteristics are NOT correlated with the regressors.
- If we reject the null, we are rejecting the statistical assumptions behind RE models.

Hausman Tests

- Prefer RE to FE estimates unless the Hausman test is rejected.

  $\rightarrow$ Rejection suggests that unobserved time-invariant factors are correlated with some explanatory variable.

  $\rightarrow$ Failure to reject means RE and FE are sufficiently close that it doesn't matter which is used, so RE preferred because it's more efficient.

Example: Haviv, Huang & Li's (2020) work on video game platforms.

Haviv, Huang & Li (2020)

Does selling more games on a console this month raise the sales of the other games on that console next month?

Haviv, Huang & Li (2020)

- Monthly US sales, January 2007–December 2012, from the NPD Group. The two consoles are the Xbox 360 and the PlayStation 3.
- The outcome is the log of copies sold of game j on console k in month t.
- 605 games appear on both consoles. 508 are console-exclusives.
- Do exclusives lift sales on OTHER games on the same console?

The panel

- The outcome is the log of copies sold of game j on console k in month t.

What they find

- A 1% increase in copies sold on a console raises the sales of the other games on that console next month by 0.153% (standard error 0.043).

Example

- *Gears of War 3* is exclusive to the Xbox 360. It sold 1.8 million copies in September 2011, about 30% of Xbox game sales that month.
- That month of sales raised other Xbox games by 4.6% the next month, about 219,000 extra copies.
- They read the spillover as habit. Finishing a game leaves the same owners ready to buy another.

Thanks for listening :)